

Week 2 Day 4 Compose App Lab

Purpose

이 실습은 Day 3에서 길게 입력한 `docker run` 옵션을 `compose.yaml` 로 옮긴다. 핵심은 파일 하나로 `web`, `db`, `network`, `volume`, `environment`, `healthcheck`, `cleanup`을 재현하는 것이다.

Setup

```
cd week2/day4/labs/compose-app
cp .env.example .env
```

`.env` 의 `POSTGRES_PASSWORD` 는 로컬 실습용 값으로 바꾼다. 실제 운영 secret이나 개인 password를 쓰지 않는다.

Run

```
docker compose config
docker compose up -d
docker compose ps
for i in 1 2 3 4 5; do
  curl -I http://localhost:18084 && break
  sleep 1
done
curl -s http://localhost:18084 | grep compose-site-v1
docker compose logs db
docker compose run --rm db-client
```

Expected Evidence

```
web-1    running
db-1     healthy
HTTP/1.1 200 OK
compose-site-v1
current_database | current_user
paperclip       | paperclip
```

Failure Drill

`.env` 에서 `POSTGRES_PASSWORD` 줄을 제거한 뒤 `config`를 확인한다.

```
docker compose config
```

기대 결과는 `set POSTGRES_PASSWORD in .env` 메시지가 포함된 실패다.

Cleanup

```
docker compose down
docker compose down -v
```

`docker compose down` 은 container와 network를 정리한다. `docker compose down -v` 는 named volume까지 삭제하므로 DB data가 사라진다.

File Map

Path	Purpose
compose.yaml	web, db, db-client service 정의
.env.example	local .env 작성 기준
html/index.html	nginx가 제공하는 정적 페이지
db/init/001_schema.sql	PostgreSQL 최초 초기화 SQL
README.md	실행, 확인, 실패, 정리 기준

Services

Service	Image	Role	Exposed To Host
web	nginx:1.27-alpine	static HTML server	yes, \${WEB_PORT:-18084}:80
db	postgres:16-alpine	PostgreSQL database	no
db-client	postgres:16-alpine	query/check tool	no, profile tool

`db-client` 는 기본 application service가 아니라 확인용 service다. 필요할 때 `docker compose run --rm db-client` 로 실행한다.

Environment Variables

Variable	Required	Default	Notes
WEB_PORT	no	18084	host에서 web에 접근할 port
POSTGRES_DB	no	paperclip	practice database name
POSTGRES_USER	no	paperclip	practice user
POSTGRES_PASSWORD	yes	none	.env에서 로컬 값으로 설정

`POSTGRES_PASSWORD` 는 required variable이다. 값이 없으면 `docker compose config` 가 실패한다. 의도적으로 실행 전 실패하게 만들어 secret 누락을 빨리 발견한다.

Recommended Local Setup

```
cp .env.example .env
```

그 다음 `.env` 를 열어 `POSTGRES_PASSWORD` 값을 로컬 실습용 값으로 바꾼다. 실제 개인 password, Docker Hub token, cloud access key를 쓰지 않는다.

Preflight

```
docker version
docker compose version
docker compose config
```

Expected:

```
services:
  db:
  web:
networks:
  app-net:
volumes:
  pgdata:
```

Detailed Runbook

1. Start

```
docker compose up -d
```

2. Check service status

```
docker compose ps
```

Expected signals:

```
web   Up
db    Up (healthy)
0.0.0.0:18084->80/tcp
```

3. Check web

```
curl -I http://localhost:18084
curl -s http://localhost:18084 | grep compose-site-v1
```

Expected:

```
HTTP/1.1 200 OK
compose-site-v1
```

4. Check DB

```
docker compose logs db
docker compose run --rm db-client
```

Expected:

```
database system is ready to accept connections
current_database | current_user
paperclip        | paperclip
```

Network Model

The host reaches the web service through published port `localhost:18084` .

Containers inside the Compose network reach PostgreSQL through service name `db` .

From	To	Address
host browser/curl	web	localhost:18084
db-client container	db	db:5432
host terminal	db	not published by default

Do not document the DB container IP as the connection contract. Container IPs can change. Use service name `db` .

Volume Model

`pgdata` stores PostgreSQL data at `/var/lib/postgresql/data` .

Command	Result
<code>docker compose down</code>	removes containers and network
<code>docker compose down -v</code>	also removes named volumes

Use `down -v` only when resetting practice data. Do not use it as the default cleanup for data that must be preserved.

Failure Drill 1: Missing Env

Remove `POSTGRES_PASSWORD` from `.env` .

```
docker compose config
```

Expected failure:

```
set POSTGRES_PASSWORD in .env
```

Fix:

```
cp .env.example .env
```

Then set a local practice password again.

Failure Drill 2: Wrong Port

```
curl -I http://localhost:80
curl -I http://localhost:18084
docker compose ps
```

Interpretation:

- `localhost:80` may fail because the published host port is `18084` .
- `localhost:18084` should return HTTP 200 when web is running.
- This is a host port issue, not an nginx container port issue.

Failure Drill 3: Stale Volume

If you edit `db/init/001_schema.sql` after the database has already initialized, the change may not appear. PostgreSQL init scripts run only during initial database creation.

Practice reset:

```
docker compose down -v
docker compose up -d
```

Warning: this deletes practice DB data.

Short RCA Template

Compose App RCA

- Symptom:
- Failed command:
- Error excerpt:
- Category: config / port / env / network / volume / readiness
- Hypothesis:
- Fix:
- Recheck:
- Prevention:

Security Notes

- Do not commit `.env`.
- Do not put real passwords or tokens in `.env.example`.
- Do not paste Docker Hub tokens, MFA codes, cloud access keys, or personal credentials into terminal screenshots.
- Prefer placeholder values in documentation.
- Keep DB port unpublished unless the exercise specifically requires host DB access.

Handoff Section For Student README

Docker Compose

Setup

```
```bash
cp .env.example .env
```
```

Run

```
```bash
docker compose config
docker compose up -d
```
```

Check

```
```bash
docker compose ps
curl -I http://localhost:18084
curl -s http://localhost:18084 | grep compose-site-v1
docker compose run --rm db-client
```
```

Cleanup

```
```bash
docker compose down
```
```

Reset practice DB data:

```
```bash
docker compose down -v
```
```

Evidence Checklist

- ☐ `docker compose config` succeeded.
- ☐ `docker compose ps` shows web running.
- ☐ `docker compose ps` shows db running or healthy.
- ☐ `curl -I http://localhost:18084` returns HTTP 200.
- ☐ body contains `compose-site-v1`.
- ☐ `db-client` query returns `paperclip`.
- ☐ missing env failure is documented.
- ☐ cleanup command is documented.
- ☐ `down -v` data deletion warning is documented.

Scoring

| Item | 0 | 1 | 2 |
|----------|----------------|-------------------|--|
| Config | not run | run only | output interpreted |
| Web | no check | status only | HTTP and body marker |
| DB | no check | healthy only | query evidence |
| Network | no explanation | network name only | service name db explained |
| Volume | no note | volume listed | data lifecycle explained |
| Security | secret exposed | warning only | <code>.env.example</code> and no secret exposure |
| Handoff | none | commands only | expected results and cleanup warning |

Completion Statement

This Compose app is complete when another student can clone the files, create `.env`, run `docker compose up -d`, verify web and DB evidence, and clean up without accidentally deleting data they intended to keep.